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10/02/2024

Code-A_(Phase-1)

Aakash + BYJU'S

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Ph.011-47623456

MM : 720 NCERT Booster Test Series-RM(P1)2324-T04A **Time : 200 Min.**

Topics Covered:
Physics : Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves
Chemistry : Solutions, Chemical Kinetics, Organic Chemistry- Some Basic Principles and Techniques
Botany : Photosynthesis in higher plants, Respiration in plants, Plant Growth and Development
Zoology : Chemical Coordination and Integration, Animal Kingdom, Structural Organisation in Animals– Animal Morphology (Cockroach, Frog, Earthworm)

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. In an isothermal expansion, change in internal energy and work done by the gas are respectively
   (1) Positive, zero
   (2) Zero, positive
   (3) Positive, negative
   (4) Zero, zero

2. At 20°C the value of the density of a fixed mass of an ideal gas divided by its pressure is 'k'. At 120°C, this ratio will be equal to
   (1) k
   (2) $\frac{293}{393}k$
   (3) $\frac{20}{120}k$
   (4) $\frac{393}{293}k$

3. A particle executing SHM with angular frequency $\frac{\pi}{2}$ rad/s. The time taken by particle to move from mean position to extreme position is
   (1) 4 seconds
   (2) 2 seconds
   (3) 8 seconds
   (4) 1 seconds

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4. A plane progressive wave is given as $y = A_0 \sin\left(\frac{2\pi x}{\lambda} - \omega_0 t + \frac{\pi}{2}\right)$. If magnitude of maximum particle velocity is two times as that of wave velocity then wavelength of the wave is (Symbols have usual meanings)
(1) πA_0
(2) $2\pi A_0$
(3) $5\pi A_0$
(4) $4\pi A_0$

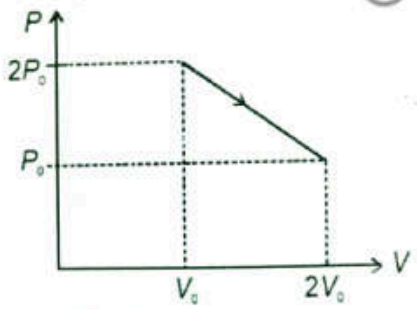
5. For a particle which is undergoing simple harmonic motion, the square of velocity v^2 is plotted against the square of displacement x^2. The curve will be
(1) An ellipse
(2) A straight line
(3) A circle
(4) A parabola

6. A particle performs simple harmonic motion about $X = 0$ with an amplitude A and time period T. The speed of the particle at $X = \frac{A}{2}$ will be
(1) $\frac{\pi A\sqrt{3}}{T}$
(2) $\frac{\pi A}{T}$
(3) $\frac{\pi A\sqrt{3}}{2T}$
(4) $\frac{3\pi^2 A}{T}$

7. The equation of stationary wave along a stretched string is given by $y = 4\sin\frac{\pi x}{6}\cos 20\pi t$, where x and y are in cm and t is in second. The wavelength of the wave is
(1) 12 cm
(2) 4 cm
(3) 3 cm
(4) 6 cm

8. If radius of a gas molecule is r, then mean free path of this molecule is proportional to
(1) r^2
(2) r
(3) $\frac{1}{r^2}$
(4) r^0

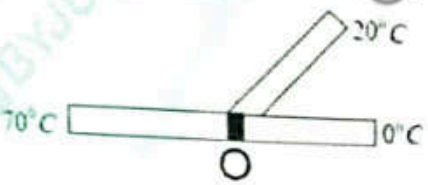
9. The work done by the gas in the process shown in given P-V diagram is



(1) P_0V_0
(2) $2P_0V_0$
(3) $\frac{3}{2}P_0V_0$
(4) $\frac{P_0V_0}{2}$

10. The equation of motion of a particle executing SHM is $a + 16\,\pi^2 x = 0$. In this equation a is particle's acceleration in m s^{-2} and x is displacement in meter. The time period of SHM is
(1) 1 s
(2) 2 s
(3) 0.5 s
(4) 3 s

11. Three rods of same dimensions and same material are arranged as shown. The junction temperature in steady state of heat conduction is



(1) 30°C
(2) 40°C
(3) 50°C
(4) 60°C

12. The ratio among the linear expansion coefficient (α), areal expansion coefficient (β) and volume expansion coefficient (γ) for a metal is
(1) 1 : 2 : 3
(2) 1 : 1 : 1
(3) 4 : 3 : 2
(4) 3 : 1 : 2

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13. For one mole of an ideal gas, the internal energy is given by $U = 3 + 1.5PV$ then gas is
(1) Monoatomic
(2) Diatomic
(3) Polyatomic
(4) Gas mixture of monoatomic and diatomic

14. The root mean square speed of four molecules of a gas having speeds 4 m/s, 5 m/s, 6 m/s and 7 m/s is about
(1) 5.6 m/s
(2) 2.4 m/s
(3) 6.2 m/s
(4) 4.8 m/s

15. Equation of SHM of a particle is given by $y = A\sin\frac{2\pi}{T}t$. Time taken by particle to move from $y = \frac{A}{2}$ to A will be
(1) $\frac{T}{12}$
(2) $\frac{T}{4}$
(3) $\frac{T}{8}$
(4) $\frac{T}{6}$

16. A gaseous mixture consists of 8 g of helium and 32 g of oxygen. The ratio (C_p/C_v) of the mixture is
(1) $\frac{11}{17}$
(2) $\frac{17}{11}$
(3) $\frac{18}{11}$
(4) $\frac{10}{9}$

17. The fastest mode of heat transfer is
(1) Conduction
(2) Radiation
(3) Convection
(4) All of these are equally fast

18. A body kept in a room and it cools from 80°C to 70°C in T_1 minute, from 70°C to 60°C in T_2 minute and from 60°C to 50°C in T_3 minutes, then
(1) $T_1 < T_2 < T_3$
(2) $T_1 = T_2 = T_3$
(3) $T_1 > T_2 > T_3$
(4) $T_1 < T_2 > T_3$

19. A string of length 1 m is fixed at both end under a tension of 10 N. If mass of string is 4 g, then its frequency of fundamental mode is
(1) 200 Hz
(2) 50 Hz
(3) 25 Hz
(4) 100 Hz

20. A heat engine is designed to work with temperatures 27^0C and 227^0C. Its efficiency may be
(1) 0.5
(2) 0.75
(3) 0.6
(4) 0.33

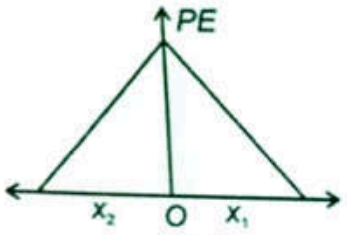
21. If $y = [\sin(\omega t) + \cos(\omega t)]$ represents displacement of particle executing SHM, then maximum acceleration of the particle is
(1) ω^2
(2) $2\omega^2$
(3) $\sqrt{2}\omega^2$
(4) ω

22. The dimensional formula of specific heat capacity is
(1) $[M^0L^2T^{-2}K^{-1}]$
(2) $[ML^2T^{-2}K]$
(3) $[M^0L^2T^{-2}K^{-2}]$
(4) $[ML^2T^{-2}K^{-2}]$

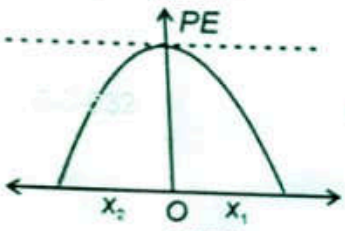
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23. A particle of mass m is executing SHM with equilibrium position O. Its potential energy is plotted against its displacement from O. The correct graph is

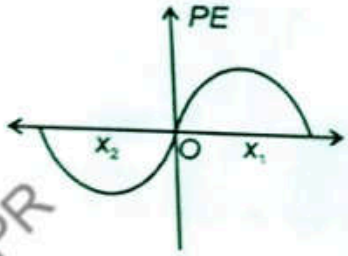
(1)



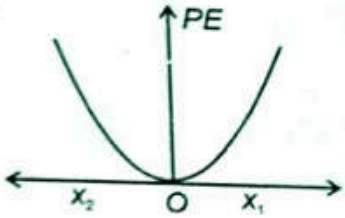
(2)



(3)



(4)



24. The Kinetic Energy and Potential Energy of a particle executing SHM with amplitude A will be equal when its displacement from mean position is

(1) $A/2$

(2) $\frac{A}{\sqrt{2}}$

(3) $\frac{\sqrt{3}A}{2}$

(4) A

25. Which among the following is a correct unit for thermal conductivity of any material?

(1) $K\,W^{-1}\,m^{-2}$

(2) $W\,m^2\,m^{-1}$

(3) $W\,K\,m^{-1}$

(4) $W\,m^{-1}\,K^{-1}$

26. An insulated container containing diatomic gas of molar mass M is moving with a velocity v_0. If container is suddenly stopped, then change in temperature is

(1) $\frac{Mv_0^2}{5R}$

(2) $\frac{5Mv_0^2}{R}$

(3) $\frac{5Mv_0^2}{2R}$

(4) $\frac{2Mv_0^2}{5R}$

27. If a gas is allowed to expand adiabatically against external pressure

(1) There is decrease in internal energy of gas

(2) Pressure remains constant

(3) Its temperature remains constant

(4) There is increase in its internal energy of gas

28. When a wave travels in a medium, the particle displacement is given by, $y = p\sin 2\pi(qt - rx)$, where p, q and r are constants. The maximum particle velocity will be twice the wave velocity if

(1) $p > qr$

(2) $q = pr$

(3) $r = \frac{1}{pr}$

(4) $r > p\pi$

29. Two tuning forks A and B have frequency 311 Hz and 310 Hz, then what will be beat frequency if both are sounded together?

(1) 2

(2) 1

(3) $\frac{1}{2}$

(4) 4

30. The vibrations of a string of length 60 cm fixed at both the ends are represented by the equation

$y = 2\sin\left(\frac{4\pi x}{15}\right)\cos\left(96\pi t\right)$ where x and y are in cm. The number of loops formed on the string is

(1) 6

(2) 16

(3) 5

(4) 15

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31. Molar specific heat capacity at constant volume, for a linear triatomic gas is (Neglect vibrational mode)

(1) $4R$

(2) $\frac{5}{2}R$

(3) $\frac{3}{2}R$

(4) $\frac{7}{2}R$

32. The first law of thermodynamics is based on

(1) Conservation of angular momentum

(2) Conservation of mass

(3) Conservation of energy

(4) Conservation of momentum

33. For an isobaric process, a gas expand from V to $4V$ then its internal energy

(1) Increases

(2) Decreases

(3) Remains unchange

(4) May increase or decreases depending on atomicity of gas

34. A uniform metal rod, having young's modulus Y, coefficient of linear expansion α is clamped horizontally, between two rigid supports on decreasing the temperature by $\Delta\theta$ thermal stress developed in the rod will be

(1) $\frac{1}{2}\frac{\alpha\Delta\theta}{Y}$

(2) $Y\alpha\Delta\theta$

(3) $\frac{Y\alpha}{\Delta\theta}$

(4) $\frac{Y\Delta\theta}{\alpha}$

35. Under which of the following conditions is the law $PV = RT$ obeyed most closely by a real gas?

(1) High pressure and high temperature

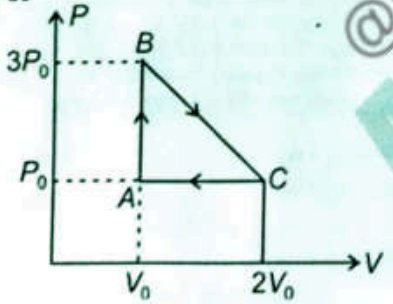
(2) Low pressure and low temperature

(3) Low pressure and high temperature

(4) High pressure and low temperature

SECTION-B

36. One mole of an ideal diatomic gas is taken round the cyclic process $ABCA$ as shown in figure. The work by the gas in the cycle will be



(1) P_0V_0

(2) $-P_0V_0$

(3) $\frac{-P_0V_0}{2}$

(4) $\frac{-3P_0V_0}{2}$

37. Find the ratio of time periods of two identical springs if they are first joined in series and then in parallel and a mass m is suspended from them

(1) 4

(2) 2

(3) 1

(4) 3

38. A : In simple harmonic motion, the graph between velocity and the displacement is a parabola.

R : In simple harmonic motion, the phase difference between velocity and displacement is π.

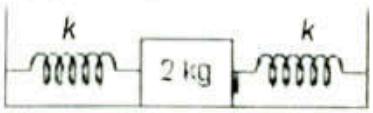
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

39. A 2 kg block is attached to two identical springs of spring constant 10 N/m as shown below. The angular frequency of system for small oscillations is



(1) $\sqrt{10}$ rad/s

(2) $\sqrt{5}$ rad/s

(3) $2\sqrt{2}$ rad/s

(4) 2 rad/s

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40. If two bodies are in thermal equilibrium, then
(1) Both bodies have same specific heat
(2) Both bodies have same internal energy
(3) Both bodies have same temperature
(4) Both bodies have same number of moles

41. If the mean free path of the gas is λ at one atmospheric pressure, then its value at 5 atmospheric pressure will be (temperature is assume to be constant)
(1) $\frac{\lambda}{5}$
(2) 5λ
(3) λ
(4) $\frac{2}{3}\lambda$

42. For an ideal gas, $R = \frac{2}{5}C_p$, where C_p is molar specific heat capacity at constant pressure. In an adiabatic process, pressure of the gas is proportional to T^n. The value of 'n' is
(1) $\frac{5}{2}$
(2) $\frac{5}{3}$
(3) $\frac{7}{5}$
(4) $\frac{7}{2}$

43. If tension and diameter of a sonometer wire having fundamental frequency n are made 4 times and density is doubled, then the new fundamental frequency becomes (Length remains same)
(1) $\frac{1}{\sqrt{2}}n$
(2) $\frac{1}{2\sqrt{2}}n$
(3) $\frac{1}{2}n$
(4) n

44. The phase difference between velocity and displacement of a particle executing SHM is
(1) 0
(2) $\frac{\pi}{4}$
(3) π
(4) $\frac{\pi}{2}$

45. Among the following, in which case, the Doppler effect would never be observed?
(1) Source is moving but the observer is stationary
(2) Observer is moving but the source is stationary
(3) Both the source and observer are moving
(4) Both the source and observer are stationary

46. Newton's formula for velocity of sound in gas is
(1) $v = \sqrt{\frac{2P}{\rho}}$
(2) $v = \frac{2}{3}\sqrt{\frac{P}{\rho}}$
(3) $v = \sqrt{\frac{\rho}{P}}$
(4) $v = \sqrt{\frac{P}{\rho}}$

47. The successive frequencies of standing waves setup in an organ pipe are 300 Hz, 450 Hz, 600 Hz. Then frequency of 5th harmonics is
(1) 1500 Hz
(2) 1350 Hz
(3) 1650 Hz
(4) 750 Hz

48. Starting with the same initial conditions, an ideal gas expands from V_1 to V_2 in 3 different ways, the work done by the gas is W_1 if the process is purely isothermal, W_2 if purely isobaric and W_3 if purely adiabatic then,
(1) $W_2 > W_1 > W_3$
(2) $W_2 > W_3 > W_1$
(3) $W_1 > W_2 > W_3$
(4) $W_1 > W_3 > W_2$

49. Ratio of rms speed of oxygen molecules to the hydrogen molecules at room temperature is
(1) 1 : 4
(2) 4 : 1
(3) 16 : 1
(4) 1 : 16

50. The value of polytropic index for a gas undergoing a thermodynamic process is zero, then the process is
(1) Adiabatic
(2) Isothermal
(3) Isochoric
(4) Isobaric

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CHEMISTRY

51. The species which is not an electrophile?

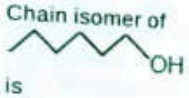
(1) $AlCl_3$

(2) $\overset{+}{Br}$

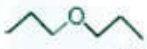
(3) $\overset{\oplus}{NH_4}$

(4) $\overset{+}{NO_2}$

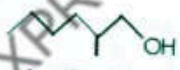
52. Chain isomer of

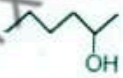


is

(1) 

(2) 

(3) 

(4) 

53. Which of the following sets represent only nucleophiles?

(1) $HS^{\ominus}$, $C_2H_5O^{\ominus}$, $(CH_3)_3N$:

(2) CH_3CHO, $H_2\overset{\ominus}{N}$:, $H_3\overset{\oplus}{O}$:

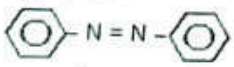
(3) BF_3, CH_3I, $H_2\ddot{O}$:

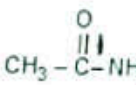
(4) $\dot{C}H_3$, $H_2\overset{\ominus}{N}$:, $\overset{\oplus}{N}O_2$

54. Kjeldahl's method is applicable to

(1) $CH_3 - NO_2$

(2) 

(3) 

(4) 

SECTION-A

55. The Violet colour developed during the Lassaigne's test for sulphur is due to

(1) $Na_4[Fe(CN)_5NOS]$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $Fe(SCN)_3$

(4) $Fe_3[Fe(CN)_6]_2$

56. Positive deviation from Raoult's law is shown by the mixture

(1) Benzene + toluene

(2) Chloroform + acetone

(3) Ethanol + acetone

(4) Phenol + aniline

57. If vapour pressure of two liquids A and B in pure state are 400 and 500 mm of Hg respectively then mole fraction of A in vapour phase of solution obtained by mixing equal moles of A and B is

(1) $\frac{1}{2}$

(2) $\frac{3}{5}$

(3) $\frac{4}{9}$

(4) $\frac{2}{3}$

58. For ideal solution which of the given relation is incorrect?

(1) $\Delta H_{mix} < 0$

(2) $\Delta S_{mix} > 0$

(3) $\Delta G_{mix} < 0$

(4) $\Delta V_{mix} = 0$

59. If 0.1 M calcium phosphate solution, dissociates by 70% then the van't Hoff factor of the solution will be

(1) 2.7

(2) 3.8

(3) 1.5

(4) 4.5

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60. Consider the following

Gas	Henry's law constant (K_H/kbar) at 293 K
He	144.97
H_2	69.16
O_2	34.86
N_2	76.48

The correct order of increasing solubility is

(1) $He < N_2 < H_2 < O_2$

(2) $O_2 < H_2 < N_2 < He$

(3) $N_2 < H_2 < He < O_2$

(4) $N_2 < He < O_2 < H_2$

61. Strongest +I effect is shown by

(1) $-CH_3$

(2) CH_3CH_2-

(3) $-\overline{O}$

(4) $(CH_3)_3C-$

62. In Duma's method for estimation of nitrogen, 0.2 g of an organic compound gave 50 mL of nitrogen collected at STP. Calculate the percentage of nitrogen in the compound.

(1) 31.25%

(2) 17.4%

(3) 20%

(4) 50%

63. How many isomers of $C_5H_{11}OH$ will be primary alcohol?(excluding stereoisomers)

(1) 2

(2) 3

(3) 4

(4) 5

64. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 1 g of the compound is neutralized by 20 mL of 0.5 M H_2SO_4. The percentage of nitrogen in the compound is

(1) 14%

(2) 28%

(3) 56%

(4) 42%

65. Total number of structural isomers of C_4H_8 is

(1) 4

(2) 5

(3) 6

(4) 3

66. **Assertion (A) :** At azeotropic point boiling point of solution is constant.
Reason (R) : Azeotropic solutions may have boiling points either greater than both the components or lesser than both the components.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

67. If 8 g of a non-electrolyte solute is dissolved in 114 g of n-octane to reduce its vapour pressure to 80%, the molar mass (in g mol^{-1}) of the solute is [Given that molar mass of n-octane is 114 g mol^{-1}]

(1) 20

(2) 40

(3) 60

(4) 80

68. Which of the following solutions has highest boiling point?

(1) 0.1 m glucose in water

(2) 0.1 m sucrose in water

(3) 0.1 m NaCl in water

(4) 0.1 m $CaCl_2$ in water

69. If the mole fraction of acetone in ethyl alcohol is 0.2 then the molality of acetone in the solution is

(1) 5.9 m

(2) 5.4 m

(3) 7.8 m

(4) 2.7 m

70. Calculate the dissociation constant of HCN if van't Hoff factor of 10^{-2} M HCN is 1.002

(1) 2×10^{-6}

(2) 4×10^{-8}

(3) 6×10^{-4}

(4) 2.5×10^{-8}

71. For $[CrCl_3 . x\ NH_3]$, elevation in boiling point of one molal solution is double of one molal solution of glucose, hence x is , if complex is 100% ionised.

(1) 3

(2) 4

(3) 5

(4) 6

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72. Assertion (A) : For a first order reaction $t_{99.9\%} = 10 \times t_{50\%}$.
Reason (R) : Half life of first order reaction is independent of initial concentration of reactants.
In the light of above statements, select the correct option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

73. Which of the following forms azeotropic mixture?
(1) 90% ethanol + 10% water
(2) 75% nitric acid + 25% water
(3) 80% ethanol + 20% water
(4) 68% nitric acid + 32% water

74. A → B, is a zero order reaction. If it takes 10 sec for complete conversion of 10^{-3} M of A into B then rate constant for the given reaction is
(1) 10^{-4} mol $L^{-1}s^{-1}$
(2) 10^{-3} mol $L^{-1}s^{-1}$
(3) 10^{-2} mol $L^{-1}s^{-1}$
(4) 10^{-1} mol $L^{-1}s^{-1}$

75. The chemical reaction, $2O_3 \longrightarrow 3O_2$ proceeds as follows:
(i) $O_3 \rightleftharpoons O_2 + O$(fast)
(ii) $O + O_3 \rightleftharpoons 2O_2$(slow)
The unit of rate constant of reaction $2O_3 \longrightarrow 3O_2$, is
(1) mol $L^{-1}s^{-1}$
(2) s^{-1}
(3) L $mol^{-1}s^{-1}$
(4) L^2 mol^{-2} s^{-1}

76. In a reaction, C + D → product, rate is tripled when the concentration of C is tripled keeping concentration of D same, and the rate increases by a factor of 9 when the concentrations of both the reactants (C and D) are tripled, rate law for the reaction can be written as
(1) Rate = k[C] $[D]^0$
(2) Rate = k[C] [D]
(3) Rate = $k[C]^1$ $[D]^2$
(4) Rate = $k[C]^2$ $[D]^2$

77. In a first order reaction the concentration of reactant decreases from 600 mol/dm^3 to 150 mol/dm^3 in 2 minutes. The rate constant of the reaction in sec^{-1} is
(1) 4.45×10^{-2}
(2) 1.15×10^{-2}
(3) 3.25×10^{-3}
(4) 2.25×10^{-2}

78. If the activation energy for the forward reaction is 80 kJ/mole and activation energy for the backward reaction is 120 kJ/mol, then enthalpy of reaction will be
(1) 40 kJ/mole
(2) −40 kJ/mole
(3) 200 kJ/mole
(4) −200 kJ/mole

79. For the reaction, 2A(g) → B(g) + C(g) + D(g) having first order kinetics. If the initial pressure is 90 mm Hg and at time 15 minutes the pressure becomes 110 mm Hg, then the pressure of A at time 15 minutes is
(1) 20 mm Hg
(2) 40 mm Hg
(3) 50 mm Hg
(4) 90 mm Hg

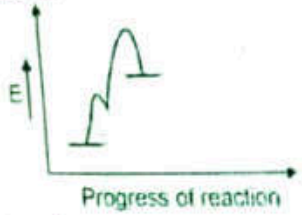
80. Consider following reaction,
$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
Initially the concentration of N_2O_5 is 3.33 mol L^{-1}. It decreases to 3.08 mol L^{-1} after 92 minutes. The rate of production of NO_2 during this period is
(1) 4.07×10^{-2} mol L^{-1} min^{-1}
(2) 6.79×10^{-4} mol L^{-1} min^{-1}
(3) 14×10^{-4} mol L^{-1} min^{-1}
(4) 5.43×10^{-3} mol L^{-1} min^{-1}

81. The differential rate law equation for given elementary reaction is
$X + 2Y \xrightarrow{k} C$
(1) $\frac{-d[X]}{dt} = \frac{-d[Y]}{dt} = \frac{d[C]}{dt} = k[X][Y]^2$
(2) $\frac{-d[X]}{dt} = -\frac{1}{2}\frac{d[Y]}{dt} = \frac{d[C]}{dt} = k[X][Y]^2$
(3) $\frac{-d[X]}{dt} = -\frac{1}{2}\frac{d[Y]}{dt} = \frac{d[C]}{dt} = k[X]^2[Y]^1$
(4) $\frac{-d[X]}{dt} = -\frac{1}{2}\frac{d[Y]}{dt} = \frac{1}{3}\frac{d[C]}{dt} = k[X][Y]^2$

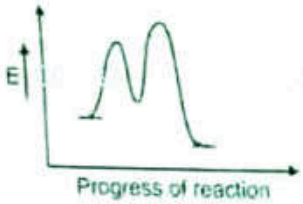
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82. For the given reaction A → B, ΔH = +xkJ
Mech. A → A* (Fast)
A* → B (Slow)
The graph corresponding to the given reaction is:

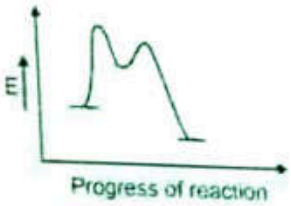
(1)



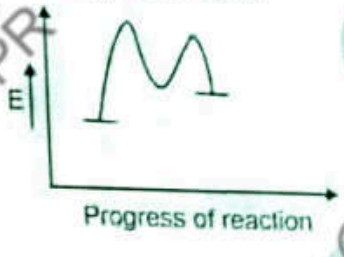
(2)



(3)



(4)



83. The rate constant for a reaction is given by
$\ln k = \frac{-2000}{T} + 20.14$
Activation energy, E_a for this reaction will be

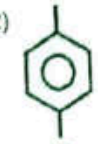
(1) 1.67 kJ mol^{-1}
(2) 4 kcal mol^{-1}
(3) 38.3 kJ mol^{-1}
(4) 9 kcal mol^{-1}

84. Example of coplanar molecule is

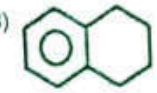
(1)



(2)



(3)



(4)



85. Which of the following will give blood red colour with Lassaigne's test for nitrogen followed by reaction with ferric ions?

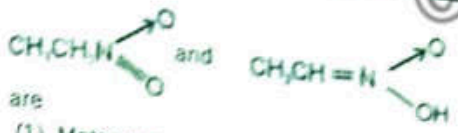
(1) $(NH_2)_2C = O$
(2) $C_6H_5SO_3H$
(3) $(NH_4)_2SO_4$
(4) 

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SECTION-B

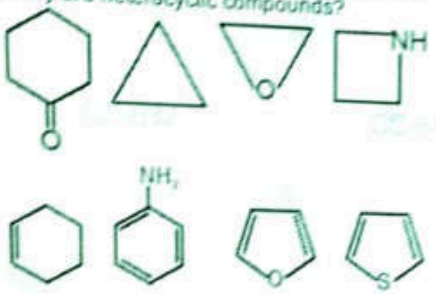
86. o-nitrophenol and p-nitrophenol are separated by

(1) Fractional distillation
(2) Steam distillation
(3) Vacuum distillation
(4) Column chromatography

87.  and are

(1) Metamers
(2) Tautomers
(3) Position isomers
(4) Geometrical isomers

88. Among the given set of compounds, how many are heterocyclic compounds?



(1) 6
(2) 5
(3) 4
(4) 8

89. Negative electromeric effect is observed in case of

(1) $CH_3CH_2OH + HCl$
(2) $CH_3Cl + NaOH$
(3) $CH_3 - CH = CH_2 + HCl$
(4) $CH_3CHO + CH_3MgBr$

90. Correct IUPAC name of the following compound is



(1) N, N-Diethylbenzenamine
(2) N-Ethyl-N-phenylethanamine
(3) N-Phenyl-N-ethylethanamine
(4) N, N-Dimethylbenzenamine

91. For given experimental data order of reaction will be

Initial concentration	1 M	2 M	4 M
Half life	5 min	10 min	20 min

(1) 0
(2) 3
(3) 1
(4) 2

92. For the reaction, $2A(g) \rightarrow B(g) + \frac{1}{2}C(g)$, the value of rate of disappearance of A(g) is given as 2×10^{-2} mol $L^{-1} s^{-1}$. The rate of formation of C and rate of reaction are given respectively as

(1) $10^{-3}\ Ms^{-1}$ and $5 \times 10^{-2}\ Ms^{-1}$
(2) $10^{-2}\ Ms^{-1}$ and $5 \times 10^{-3}\ Ms^{-1}$
(3) $5 \times 10^{-3}\ Ms^{-1}$ and $10^{-2}\ Ms^{-1}$
(4) $2.5 \times 10^{-2}\ Ms^{-1}$ and $2 \times 10^{-2}\ Ms^{-1}$

93. In a zero-order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become

(1) 64 times
(2) 128 times
(3) 256 times
(4) 512 times

94. If the slope of graph log K vs $\frac{1}{T}$ is –10, then activation energy of the reaction is

(1) R
(2) 23.03R
(3) $\frac{R}{23.03}$
(4) 10R

95. Incorrect statement about catalyst is

(1) It reduces the activation energy between the reactants and products
(2) It catalyses the spontaneous as well as non-spontaneous reactions
(3) It does not alter Gibbs energy
(4) It does not change the equilibrium constant

96. The molecular weight of benzoic acid in benzene is determined by depression in the freezing point method corresponds to

(1) Tetramerisation of benzoic acid
(2) Dimerisation of benzoic acid
(3) Trimerisation of benzoic acid
(4) Ionisation of benzoic acid

97. The gas used in tanks by scuba divers for breathing contains

(1) 78% N_2, 21% O_2 and 1% He
(2) 11.7% He, 58% N_2 and 30.3% O_2
(3) 32.1% O_2 and 67.1% He
(4) 11.7% He, 56.2% N_2 and 32.1% O_2

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98. The temperature independent concentration term is
(1) Molarity
(2) Mole fraction
(3) Normality
(4) %(w/v)

99. Unit of Ebullioscopic constant is
(1) K mol kg^{-1}
(2) K $kgmol^{-1}$
(3) kg mol^{-1}
(4) kg mol g^{-1}

100. What is the correct sequence of osmotic pressure of decinormal aq. solution of following? (assume all electrolyte dissociate completely)
(I) $BaCl_2$, (π_1)
(II) $Al_2(SO_4)_3$, (π_2)
(III) NaCl, (π_3)
(IV) $AlCl_3$, (π_4)
(1) $\pi_2 > \pi_4 > \pi_1 > \pi_3$
(2) $\pi_3 > \pi_1 > \pi_4 > \pi_2$
(3) $\pi_1 > \pi_2 > \pi_3 > \pi_4$
(4) $\pi_3 > \pi_4 > \pi_1 > \pi_2$

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SECTION-A

101. Products of light reaction are
(1) O_2 and CO_2
(2) CO_2 and H_2O
(3) ATP and NADPH
(4) ATP and CO_2

102. The experiment performed by Jan Ingenhousz concludes that
(1) Plant restores air whatever breathing animals and burning candles remove
(2) Sunlight is essential for the plant process that helps to somehow purify the air fouled by burning candles
(3) Glucose is produced when plants grow and glucose is usually stored as starch
(4) Oxygen evolved by the green plants comes from H_2O, not from carbon dioxide

103. In C_4 plants, Calvin cycle occurs in
(1) Cytosol of mesophyll cell
(2) Chloroplast of mesophyll cell
(3) Cytosol of bundle sheath cell
(4) Chloroplast of bundle sheath cell

104. The site of light reaction is
(1) Palade particle
(2) Cisternae of golgi
(3) Stroma of chloroplast
(4) Thylakoid

105. Cyclic photophosphorylation differs from non-cyclic photophosphorylation as the former
(1) Synthesises both ATP and NADPH
(2) Does not require external source of electrons
(3) Is associated with the photolysis of water
(4) Involves two different photosystems

106. Accessory pigments
(1) Absorb light of different wavelength and prevent photo-oxidation of chlorophyll a
(2) Include all pigments other than chlorophyll *b*
(3) Are called antenna molecules which trap energy from chlorophyll molecules
(4) Can absorb only narrow range of wavelength of visible spectrum

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107. According to chemiosmotic hypothesis, ATP synthesis requires all, **except**
(1) Electron gradient
(2) Proton gradient
(3) Proton pump
(4) ATP synthase

108. Select the **incorrect** statement w.r.t. photosynthesis.
(1) CO_2 is the major limiting factor for photosynthesis
(2) The dark reactions being enzymatic are temperature controlled
(3) The primary CO_2 acceptor in the Calvin cycle is a 2-carbon compound, so the first product is a 3-C acid
(4) Photosynthesis is a physico-chemical process in which green plants use light energy to drive synthesis of organic compounds

109. At low light intensities, increase of CO_2 concentration will
(1) Increase the rate of photosynthesis in C_3 plants only
(2) Increase the rate of photosynthesis in C_4 plants only
(3) Increase the rate of photosynthesis in both C_3 and C_4 plants
(4) Neither affect rate of photosynthesis in C_3 plants nor in C_4 plants

110. Which statement is **not** correct w.r.t. glycolysis?
(1) Glycolysis occurs in cytosol
(2) Pyruvic acid is the end product
(3) Two ATP molecules are utilized for each glucose molecule
(4) O_2 is consumed and CO_2 is released

111. Complex IV of mitochondria involved in ETS is associated with all, **except**
(1) Cytochrome *c* oxidase
(2) Cytochrome *a* and a_3
(3) Cytochrome bc_1
(4) Two copper centres

112. Consider the following Assertion (A) And Reason (R) and choose the **correct** option.
Assertion (A): Carboxylation is the most crucial step of Calvin cycle.
Reason (R) : This reaction is catalysed by the enzyme RuBP carboxylase oxygenase which results in the formation of two molecules of 3-PGA.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

113. The first member of citric acid cycle is
(1) Oxaloacetic acid
(2) Citric acid
(3) Acetyl CoA
(4) Pyruvic acid

114. Each molecule of acetyl CoA produces how many NADH + H^+ and $FADH_2$ in the citric acid cycle?
(1) 4 NADH + H^+ and 1 $FADH_2$
(2) 3 NADH + H^+ and 1 $FADH_2$
(3) 8 NADH + H^+ and 2 $FADH_2$
(4) 5 NADH + H^+ and 1 $FADH_2$

115. Decarboxylation occurs during which of the following conversion steps?
(1) Pyruvic acid → Acetyl CoA
(2) Malic acid → Oxaloacetic acid
(3) Succinic acid → Fumaric acid
(4) Oxaloacetic acid → Citric acid

116. The F_1 head piece in mitochondria is a
(1) Catalytic site which produces ADP from ATP
(2) Channel through which protons cross the outer membrane
(3) integral membrane protein complex
(4) Peripheral membrane protein complex

117. Mark the respiratory substrate that has lowest RQ value among the following
(1) Fatty acid
(2) Protein
(3) Carbohydrate
(4) Malic acid

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118. Photolysis of water releases protons and oxygen. These protons get accumulated in
(1) Stroma of the chloroplast
(2) Plasma membrane of the cell
(3) Lumen of the thylakoids
(4) Space between outer and inner membranes of chloroplast

119. Direct gain of ATP through one citric acid cycle is
(1) 1 molecule
(2) 12 molecules
(3) 2 molecules
(4) 24 molecules

120. Photorespiration is a wasteful process because
(1) CO_2 and O_2 both compete for binding with the enzyme RuBisCO
(2) There is neither synthesis of ATP nor NADPH
(3) O_2 is used only in mitochondria
(4) It can occur only in the presence of light

121. EMP pathway involves
(1) Oxidative decarboxylation of pyruvate
(2) Complete oxidation of NADH
(3) Partial oxidation of glucose
(4) Decarboxylation of monosaccharides

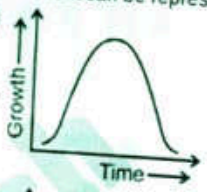
122. Amino acids can enter the respiratory pathway usually after being converted into
(1) Proteases
(2) Pyruvic acid
(3) Glycerol
(4) Peptones

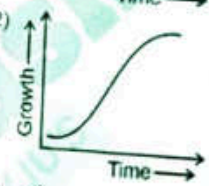
123. The final hydrogen acceptor in electron transport system of mitochondria is
(1) Cytochrome *a*
(2) ADP
(3) Cytochrome *c*
(4) Oxygen

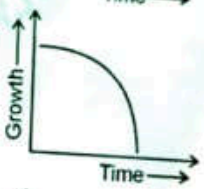
124. During the conversion of 3-phosphoglyceraldehyde to 1, 3-bisphosphoglycerate in glycolysis
(1) Energy of ATP is used
(2) Two molecules of water is released
(3) ADP is converted into ATP
(4) NADH + H^+ is formed

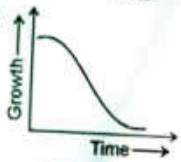
125. E. Kurosawa found that 'foolish seedling' disease in rice plant
(1) Was caused by a fungus *Gibberella fujikuroi*
(2) Occurred due to high internal concentration of auxins
(3) Occurred due to high concentration of ABA
(4) Is induced by a hormone called cytokinin

126. Which plant growth regulator does **not** occur naturally in plants?
(1) Indole butyric acid
(2) Kinetin
(3) Abscisic acid
(4) Ethylene

127. The growth curve which is characteristic of living organisms growing in a natural environment can be represented as
(1)


(2)


(3)


(4)


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128. Read the following statements and state **True** (T) or **False** (F).
A. Relative growth rate is the measurement and comparison of total growth per unit time.
B. Primary growth is characteristic of all types of plants.
C. Auxin, ethylene and ABA in general are the plant growth inhibitors.
D. Cytokinins delays cell growth while auxin is involved only in cell division.

	A	B	C	D
(1)	T	T	F	F
(2)	F	T	F	F
(3)	F	F	T	F
(4)	F	T	F	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

129. Match Column I with II and choose the **correct** option.

	Column I		Column II
a.	Ethylene	(i)	Promotes synthesis of chloroplast
b.	Auxin	(ii)	Promotes bolting
c.	Cytokinin	(iii)	Xylem differentiation
d.	Gibberellin	(iv)	Horizontal growth of seedlings.

(1) a(iv), b(i), c(iii), d(ii)
(2) a(ii), b(i), c(iii), d(iv)
(3) a(ii), b(i), c(iv), d(iii)
(4) a(iv), b(iii), c(i), d(ii)

130. Vernalisation is the process of
(1) Inducing flowering by a flash of light
(2) Increasing internodal length
(3) Promotion of flowering by giving a period of low temperature
(4) Accelerating the ability of plant to produce flower by heat treatment

131. Read the below given assertion and reason and select the correct option.
Assertion (A) : Plants exhibit plasticity in the development.
Reason (R) : Development does not lead to senescence and death.
In the light of above statements, select the **correct** option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

132. Select the **wrong** statement w.r.t ethylene
(1) Inhibits senescence and abscission
(2) Breaks seed and bud dormancy
(3) Induces flowering in mango
(4) Accelerates abscission in flowers and fruits like cotton, cherry and walnut

133. Auxin
(1) Was discovered by Skoog and Miller
(2) Counteracts apical dominance
(3) Is synthesised from precursor methionine
(4) Is widely used as weedicide

134. In most situations, ABA acts as an antagonist to which of the given plant hormones?
(1) Auxin
(2) Cytokinin
(3) Gibberellins
(4) Ethylene

135. Read the following statements.
__A__ promotes dormancy in seeds.
Malting process in brewing industry can be speed up by using __B__.
Select the **correct** option for A & B.
(1) A - Abscisic acid
B – Auxin
(2) A - Cytokinin
B - Gibberellins
(3) A - cytokinin
B – Auxin
(4) A - Abscisic acid
B - Gibberellins

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SECTION-B

136. First action spectrum of photosynthesis was described through the experiments conducted by

(1) Julius von Sachs
(2) Jan Ingenhousz
(3) C.V. Niel
(4) T.W. Engelmann

137. RuBisCO enzyme is present in all, **except**

(1) Mesophyll cells of C_3 plants
(2) Mesophyll cells of C_4 plants
(3) Bundle sheath cells of C_4 plants
(4) In CAM plants

138. The total number of ATP and NADPH required to fix six molecules of CO_2 in C_3 plants are

	ATP	NADPH
(1)	12	12
(2)	5	3
(3)	18	12
(4)	12	18

(1) (1)
(2) (2)
(3) (3)
(4) (4)

139. Read the following statements and select the **correct** option.
Statement A : Light saturation for CO_2 fixation occurs at 10% of full sunlight.
Statement B : Tomato is a greenhouse crop which can be grown in CO_2 enriched atmosphere for higher yield.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

140. Z-scheme of photophosphorylation does **not** involve

(1) Splitting of water
(2) ATP synthesis
(3) Cyclic flow of electrons
(4) $NADP^+$ reduction

141. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Phosphoglycolate	(i)	Primary CO_2 acceptor in C_3 plants
b.	Phosphoenol pyruvate	(ii)	2 carbon compound
c.	RuBisCO	(iii)	Primary CO_2 acceptor in C_4 plants
d.	RuBP	(iv)	Most abundant enzyme on earth

(1) a(iii), b(iv), c(i), d(ii)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(ii), b(iv), c(i), d(iii)
(4) a(iii), b(ii), c(iv), d(i)

142. Which of the following is **not** common between C_3 and C_4 plants?

(1) Calvin cycle
(2) Carbon fixation
(3) Krebs cycle
(4) Photorespiration

143. Respiratory pathway would be better to consider as an amphibolic pathway because

(1) In this pathway, the intermediates do not form any link with breakdown of proteins and fats
(2) Its end products are not any organic molecules
(3) It involves the synthesis of organic products from inorganic substrates
(4) It is involved in both anabolism and catabolism

144. Complete the following reaction and select the **correct** option for A, B and C.

Pyruvic acid + A + NAD^+ $\xrightarrow[B]{Mg^{2+}}$ Acetyl CoA + C + NADH + H^+

	A	B	C
(1)	CO_2	Pyruvate	O_2
(2)	CoA	Pyruvate dehydrogenase	CO_2
(3)	Lactic acid	Lactate dehydrogenase	CO_2
(4)	OAA	Citrate synthase	H_2O

(1) (1)
(2) (2)
(3) (3)
(4) (4)

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145. Each living cells in a tree is located quite close to the surface in its
(1) Woody stem
(2) Thick roots
(3) Leaves
(4) Fruits

146. Select the **incorrect** statement w.r.t. fermentation.
(1) Involves complete breakdown of glucose
(2) Oxygen is not required
(3) End products can be lactic acid or ethanol
(4) There is a net gain of 2 ATP for each molecule of glucose fermented

147. _____ is called antitranspirant hormone as it stimulates the closure of stomata.
(1) Auxin
(2) GA
(3) ABA
(4) Cytokinin

148. Match the columns and choose the **correct** option.

	Column I		Column II
a.	Meristematic zone	(i)	Proximal to meristematic zone
b.	Maturation zone	(ii)	Cells in this region are rich in protoplasm with large conspicuous nuclei
c.	Elongation zone	(iii)	Increased vacuolation
		(iv)	Protoplasmic modification

(1) a(i), b(ii, iii), c(iv)
(2) a(ii), b(iv), c(i, iii)
(3) a(ii, iii), b(i), c(iv)
(4) a(iv), b(iii, i), c(ii)

149. Which of the given plant hormones induces seedless tomato formation?
(1) Auxin
(2) Cytokinin
(3) ABA
(4) GA_3

150. If exposure of light above critical photoperiod is given to a short day plant (SDP), there will be
(1) No flowering
(2) Excessive flowering
(3) Flowering before vegetative growth is completed
(4) Fruit formation without fertilisation

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ZOOLOGY

SECTION-A

151. Hypothalamus contains several groups of neurosecretory cells called X which produce Y.
Select the option that **correctly** identifies X and Y in the above statement.
(1) X – Follicular cells, Y – Hormones
(2) X – Axons, Y – Neurotransmitters
(3) X – Nuclei, Y – Hormones
(4) X – Nuclei, Y – Neurotransmitters

152. The hormone which stimulates secretion of water and bicarbonate ions from exocrine pancreas in humans is
(1) Gastrin
(2) Secretin
(3) Cholecystokinin
(4) GIP

153. Select the **incorrect** statement w.r.t. parathyroid gland and its secretion.
(1) PTH is a hypercalcemic hormone and can maintain Ca^{2+} homeostasis.
(2) Parathyroid hormone is a steroid hormone that gets activated when there is low Ca^{2+} level in blood.
(3) PTH may lead to activation of cells responsible for dissolution of bones.
(4) PTH stimulates reabsorption of Ca^{2+} by the renal tubules.

154. The flap of connective tissue that connects the two lobes of thyroid gland is called
(1) Infundibulum
(2) Islet of Langerhans
(3) Isthmus
(4) Intercalated disc

155. Thyroxine, Erythropoietin, Cortisol, Testosterone
How many of hormones given in the box above are involved in erythropoiesis?
Select the **correct** option.
(1) 1
(2) 4
(3) 2
(4) 3

156. Match column I with column II and select the **correct** option w.r.t. antagonism.

Column I	Column II
a. Insulin	(i) Glucagon
b. Calcitonin	(ii) Parathormone
c. Aldosterone	(iii) Somatostatin
d. GHRH	(iv) Atrial natriuretic factor

(1) a(i), b(iv), c(ii), d(iii)
(2) a(i), b(iii), c(iv), d(ii)
(3) a(ii), b(i), c(iv), d(iii)
(4) a(i), b(ii), c(iv), d(iii)

157. The hormone whose hypersecretion is associated with Graves' disease is involved in
(1) Regulating the blood calcium levels
(2) Maintaining the normal rhythms of sleep-wake cycle
(3) Metabolism of carbohydrates, proteins and fats
(4) Differentiation of T-lymphocytes

158. Epinephrine and nor-epinephrine are also known as
(1) Antidiabetic hormone
(2) Gametogenic hormone
(3) Hypoglycemic hormone
(4) Emergency hormones

159. Generation of second messengers is associated with the mechanism of action of hormones
(1) Estrogen and ACTH
(2) Insulin and TSH
(3) Cortisol and ACTH
(4) Progesterone and androgens

160. While pituitary dwarfs and cretins are somewhat of same height, the main difference is that
(1) Head of cretin is larger
(2) Cretin dwarfs have long extremities
(3) Pituitary dwarfs have normal intelligence while cretins do not
(4) Pituitary dwarfs are with dark complexion

161. Read the following statements A and B and choose the **correct** option.
Statement-A: In males, LH and estrogen regulate spermatogenesis.
Statement-B: In females, only FSH induces ovulation of fully mature Graafian follicles.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

162. Cells present in testes which are responsible for secretion of hormone testosterone are
(1) Leydig cells
(2) Sertoli cells
(3) Sustentacular cells
(4) Spermatozoa

163. A person passes excessive urine and drinks much water but his glucose level in blood is normal. It is due to
(1) Increased secretion of glucagon
(2) Fall in somatostatin level in blood
(3) Increase in insulin secretion
(4) Reduction in vasopressin secretion

164. Bilateral symmetry is exhibited by all of the following animals, **except**
(1) Adult echinoderms
(2) Hemichordates
(3) Platyhelminths
(4) Chordates

165. *Nereis* can be characterised by all of the following features, **except**
(1) Aquatic annelid with parapodia
(2) Dioecious animal
(3) Presence of metameres
(4) Presence of open circulatory system

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Select the group of animals among options given below where the body cavity is **not** lined by mesoderm, instead, mesoderm is present as scattered pouches in between ectoderm and endoderm.

(1) Tapeworm, Liver fluke and Hookworm

(2) Roundworm, Filarial worm and Hookworm

(3) Earthworm, Roundworm and Tapeworm

(4) Hookworm, Roundworm and Earthworm

167. Select the animal from the options given below, whose body is cylindrical and is divided into proboscis, collar and trunk.

(1) *Saccoglossus*

(2) *Ascidia*

(3) *Petromyzon*

(4) *Myxine*

168. Presence of which one is **not** a feature of *Adamsia*?

(1) Metagenesis

(2) Gastrovascular cavity

(3) Two germ layers

(4) Cnidoblast

169. The space between the hump and ___A___ is called mantle cavity where feather-like ___B___ are present.

Choose the option which fills the blanks A and B **correctly**.

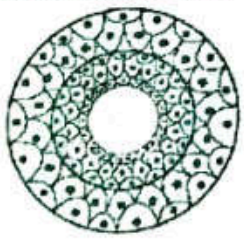
(1) **A** – Shell, **B** – Foot

(2) **A** – Mantle, **B** – Gills

(3) **A** – Gills, **B** – Mantle

(4) **A** – Mantle, **B** – Radula

170. Carefully observe the diagram and select the **correct** option w.r.t it.



(1) Pseudocoelomate, triploblastic

(2) Acoelomate, triploblastic

(3) Coelomate, diploblastic

(4) Acoelomate, diploblastic

171. Teeth are modified placoid scales and are backwardly directed in

(1) *Trygon*

(2) *Exocoetus*

(3) *Labeo*

(4) *Hippocampus*

172. Choose the **odd** one w.r.t. absence of notochord in adult stage.

(1) *Ascidia*

(2) *Salpa*

(3) *Doliolum*

(4) *Branchiostoma*

173. Which of the following features is shared by both *Hemidactylus* and *Psittacula*?

(1) Completely four chambered heart

(2) Pneumatic bones

(3) Oviparous and show direct development

(4) Poikilotherms

174. All of the following animals are included in the division Gnathostomata, **except**

(1) Hag fish

(2) Saw fish

(3) Dog fish

(4) Angel fish

175. Marine habitat, ventral mouth, absence of operculum, placoid scales, external fertilization, viviparity

How many of the above mentioned characters are associated with the most of the members of the class Chondrichthyes? Select the **correct** option.

(1) 3

(2) 4

(3) 5

(4) 6

176. Select the set of limbless animals.

(1) *Ichthyophis* and *Vipera*

(2) *Salamandra* and *Ichthyophis*

(3) *Bufo* and *Naja*

(4) *Hyla* and *Bangarus*

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177. During hibernation and aestivation, gaseous exchange in frogs take place through (A) and (B) respectively. Choose the option which represents (A) and (B) **correctly**.

	(A)	(B)
(1)	Skin	Lungs
(2)	Lungs	Skin
(3)	Buccal cavity	Buccal cavity
(4)	Skin	Skin

(1) (1)
(2) (2)
(3) (3)
(4) (4)

178. Cloaca is a small, median chamber that is used to pass all of the following to exterior, **except**
(1) Faecal matter
(2) Saliva
(3) Urine
(4) Sperms

179. Which of the following is not **correct** statement w.r.t. *Rana tigrina*?
(1) Special venous connection between liver and intestine is known as hepatic portal system.
(2) Lymph lacks few proteins and RBCs
(3) Special venous connection between kidney and upper parts of the body is known as renal portal system
(4) The ventricle opens into a sac-like conus arteriosus on the ventral side of the heart

180. Compared to that of *Homo sapiens*, the erythrocytes present in male *Rana tigrina* are
(1) Nucleated and contain haemoglobin
(2) Enucleated and contain haemoglobin
(3) Nucleated and without haemoglobin
(4) Enucleated and without haemoglobin

181. How many of the structures given in the box below are **not unpaired** structures present in *Rana tigrina*?

Kidneys, Cloaca, Liver, Optic lobes, Diencephalon, Testes

Select the correct option.
(1) Six
(2) Three
(3) Five
(4) Four

182. Read the statements given below and choose the option which states them true (T) or false (F) **correctly**.
a. All pituitary hormones are non steroidal.
b. All ovarian hormones are steroidal.
c. Cortisol acts through receptors bound to the cell membrane.
d. Thyrocalcitonin is a proteinaceous hormone released by the thyroid gland.
Select the **correct** option.
(1) a(T), b(T), c(F), d(T)
(2) a(T), b(F), c(F), d(T)
(3) a(F), b(T), c(T), d(F)
(4) a(F), b(F), c(T), d(F)

183. In male cockroach, mushroom-shaped gland is present in which of the following abdominal segments?
(1) $3^{rd} - 4^{th}$
(2) $4^{th} - 5^{th}$
(3) $5^{th} - 6^{th}$
(4) $6^{th} - 7^{th}$

184. The medulla oblongata of frog passes out through the _____ and continues into spinal cord which is enclosed in the vertebral column. Choose the option that correctly fills the blank.
(1) Cloacal aperture
(2) Foramen magnum
(3) Sensory papillae
(4) Tympanum

185. Frog's eyes are protected while in water by
(1) Retina
(2) Photoreceptor cells
(3) Nictitating membrane
(4) Skin

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SECTION-B

186. In how many of the organisms given in the box below, respiration takes place through gills?

Saccoglossus, Ophiura, Laccifer, Loligo, Pterophyllum, Alligator

Select the **correct** option.

(1) Three
(2) Four
(3) Two
(4) Five

187. Select the set of organisms which have the property of emitting light.

(1) *Pleurobrachia* and *Ctenoplana*
(2) *Ctenoplana* and *Ascaris*
(3) Liver fluke and Portuguese man-of-war
(4) *Asterias* and *Pheretima*

188. To which classes do the following animals belong? Select the **correct** option.

	Myxine	*Scoliodon*	*Exocoetus*
(1)	Chondrichthyes	Osteichthyes	Cyclostomata
(2)	Osteichthyes	Chondrichthyes	Cyclostomata
(3)	Cyclostomata	Osteichthyes	Chondrichthyes
(4)	Cyclostomata	Chondrichthyes	Osteichthyes

(1) (1)
(2) (2)
(3) (3)
(4) (4)

189. Which of the following animals lacks ear pinna?

(1) *Calotes*
(2) *Pteropus*
(3) *Rattus*
(4) *Camelus*

190. A person is suffering from nervousness, sweating, high level of blood sugar, high blood pressure. These symptoms refer to release of catecholamines from

(1) Thyroid gland
(2) Adrenal medulla
(3) β-cells of pancreas
(4) α-cells of pancreas

191. Hormone that stimulates formation of alveoli in mammary glands is

(1) FSH
(2) LH
(3) Progesterone
(4) Estrogen

192. Read the following statements and select the **correct** option.

Statement-I: Cortisol is given in the treatment of autoimmune disorders.

Statement-II: Cortisol enhances the functioning of immune system by increasing the number of B and T lymphocytes.

(1) Both statements I and II are correct
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statements I and II are incorrect

193. The sella tursica of sphenoid bone houses 'X' gland. Identify 'X' and select the **correct** option

(1) Pituitary
(2) Adrenal
(3) Pineal
(4) Ovary

194. Which hormone primarily regulates the concentration of sodium and potassium ions in the blood and other body fluids?

(1) Parathormone
(2) Aldosterone
(3) Thyrocalcitonin
(4) Adrenaline

195. **Assertion (A)** : If we cut the nerves supplying the heart of frog, it will stop beating.

Reason (R) : The heart of frog is neurogenic.

In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion
(3) Assertion is true but Reason is false
(4) Both Assertion and Reason are false

196. Select the **incorrect** statement w.r.t. the nervous system of frogs.

(1) There are 10 pairs of cranial nerves arising from the brain.
(2) The midbrain is characterised by a pair of olfactory lobes.
(3) The brain is enclosed in cranium.
(4) Hind brain consists of cerebellum and medulla oblongata.

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197. Nitrogenous excretory products of adult and larva of frog are _____ respectively.
Choose the option which fills the blank **correctly**.

(1) Urea and ammonia
(2) Ammonia and urea
(3) Ammonia and uric acid
(4) Urea and uric acid

198. Match column I and column II w.r.t. cockroach. Choose the **correct** option.

Column I		Column II
a. Hepatic caecae	(i)	6-8 in number
b. Malpighian tubules	(ii)	100-150 in number
c. Spiracles	(iii)	20 in number
d. Ommatidia in each eye	(iv)	2000 in number

(1) a(ii), b(iii), c(iv), d(i)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(ii), c(iii), d(i)
(4) a(iv), b(i), c(ii), d(iii)

199. The development of *Periplaneta americana* is through nymphal stages. Such type of development is called

(1) Holometabolous
(2) Paurometabolous
(3) Ametabolous
(4) Hemimetabolous

200. Which of the following is mismatched w.r.t. *Pheretima*?

(1) Clitellum – 14th-16th segments in mature earthworms
(2) Female genital pore – Mid-ventral line of 14th segment
(3) Pair of male genital pores – Dorso-lateral sides of the 18th segment
(4) Setae – Present in each segment of the body, except the first, last and clitellar segments

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